

Chemistry

CYL1010



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Contents

Title	Chemistry	Number	CYL1010
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1 st Year	Type	Compulsory
Prerequisite	Nil		
Objectives 1. Relate the fundamental principles of chemistry with practical problems encountered for engineers and place the emphasis on problem-solving 2. Provide knowledge of engineering materials metals, composites, polymers, cement, steel, fuels, batteries and drugs. Learning Outcomes 1. Understanding the behavior of matter and materials using fundamental knowledge of their nature. 2. Understanding the basic concepts and implementation of the same in advanced and interdisciplinary areas. 3. Predict potential complications from combining various chemicals or metals in an engineering setting. Contents <i>Atomic and Molecular Structure [6 Lectures]:</i> particle in a box, angular momentum, wavefunction, multielectron atoms, homonuclear and heteronuclear diatomic molecules, non-covalent interactions <i>Solid State Chemistry [6 Lectures]:</i> Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties. <i>Surface Chemistry[3 Lectures]:</i> Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension. <i>Electrochemical Systems [5 Lectures]:</i> Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion <i>Basic Spectroscopy [7 Lectures]:</i> Selection Rules, Fundamentals FTIR, Raman, UV-visible and Fluorescence spectroscopy <i>Transition Metal Chemistry [5 Lectures]:</i> Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry <i>Organic Chemistry [8 Lectures]:</i> Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies. Textbooks 1. Silberberg, M., <i>Chemistry: The Molecular Nature of Matter and Change</i>, 6th Edition, McGraw Hill Education. 2. West, A.R., (2015), <i>Solid State Chemistry and Its Applications</i>, 2nd edition, John Wiley & Sons. 3. J. D. Lee, <i>Concise Inorganic Chemistry</i>, 5th Edition. 4. Atkins, <i>Physical Chemistry</i>, 11th Edition. Reference Books 1. McMurry, J. E. & Fay, R. C. <i>Chemistry</i>, 5th Edition, Pearson. 2. Jonathan Clayden, Nick Greeves, Stuart Warren, <i>Organic Chemistry</i>, 2nd Edition. 3. F. Albert Cotton, Geoffrey Wilkinson, <i>Basic Inorganic Chemistry</i>, Wiley Student Edition. Self Learning Material 1. https://nptel.ac.in/downloads/122101001/			

Contents

- **Surface Chemistry[3 Lectures]:** Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension.
- **Solid State Chemistry [6 Lectures]:** Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties.
- **Electrochemical Systems [5 Lectures]:** Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion



INTRODUCTION

Electro chemistry is the branch of chemistry which deals with transformation of electrical energy into chemical energy vice versa.

- Electrochemistry serves as an interface between chemistry and everyday life. When we switch on calculator, a computer, a radio, start a car, use mobile phone, use a digital watch, and so on we are depending upon the electrochemical reactions occurring in the cells or batteries.

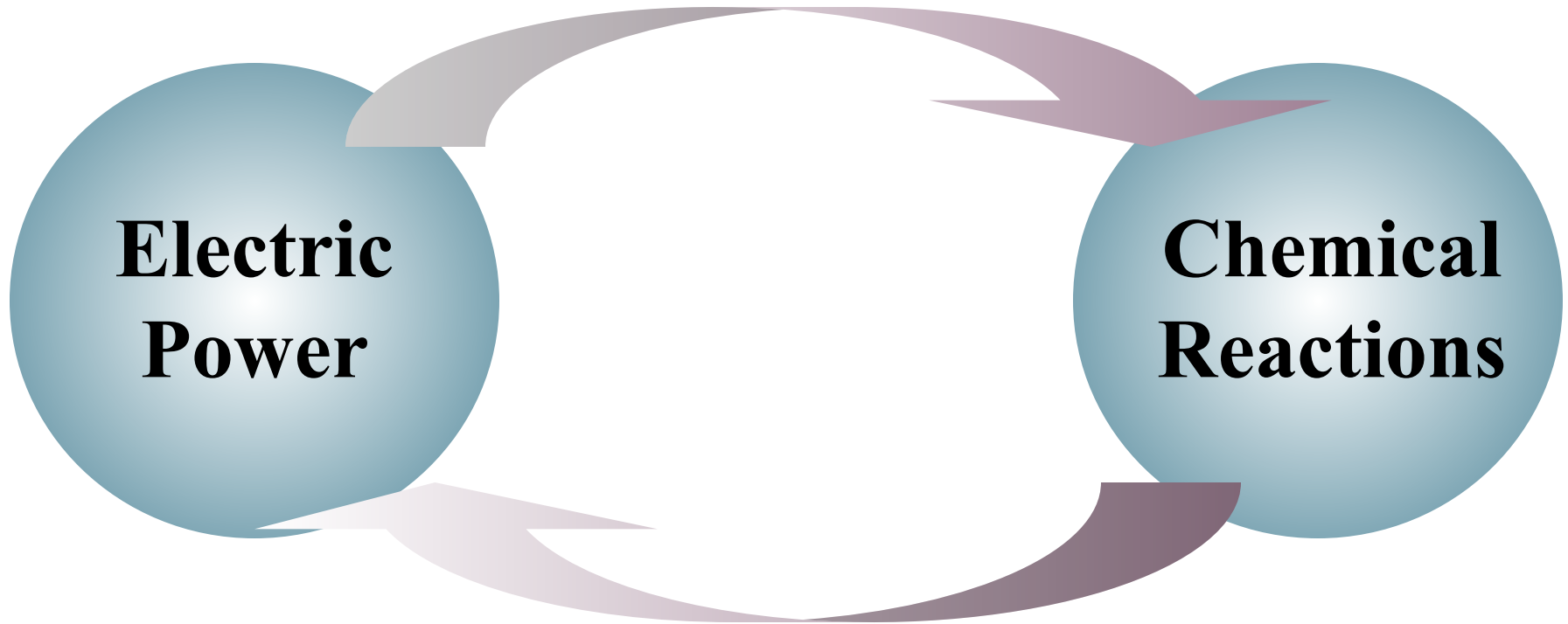
A number of Electro chemical industries have been established on the concept of electrochemistry:

- Electro refining(Cu), Electro plating(Ni, Cr, Au,Ag), Fused electrolysis(Al, Mg,Na), Electro thermal(CaC_2 , Graphite, SiC), Electrolytic furnacing (Alloy steels), Chemical to electrical Conversion (batteries, fuel cells), Gas phase electric discharge(O_3)

In addition to the above industries, the chemical Engineer uses Electro chemical principle in dealing with corrosion phenomena.

Electric power conversion in electrochemistry

Electrolysis / Power consumption



Electrochemical battery / Power generation



Electrochemistry

Electrochemistry

- Cell Potentials and Standard cell potentials, EMF
- Galvanic cells/Voltaic Cell
- Electrolytic cells
- Faraday's Law of Electrolysis
- Applications



Electrochemistry

- Much of the early work on cell physics & chemistry came from Michael Faraday - 1791 to 1867
- He was responsible for uncovering ONs, terminology (anode, cathode, etc), benzene, electromagnetism, Faraday's Laws & Faraday's constant.
- 1 Faraday = Charge on 1 mole of electrons = **96,500** c.



Electrochemistry

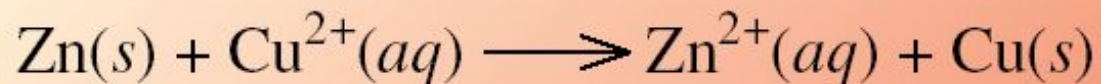
□ Redox reactions are those involving the oxidation and reduction of species.

LEO says GER...

Losing electrons □ **Oxidation**

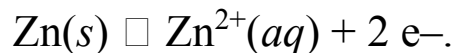
Gaining Electrons □ **Reduction**

NOTE: Oxidation and reduction *must* occur together.
They cannot exist alone.



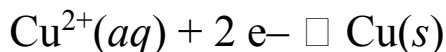
Oxidation

Half-Reaction:



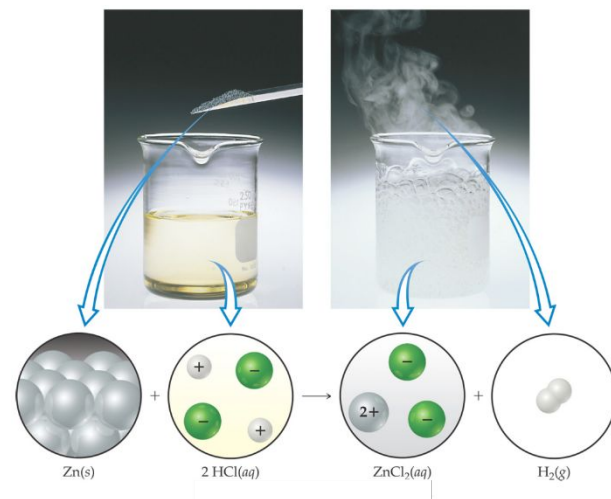
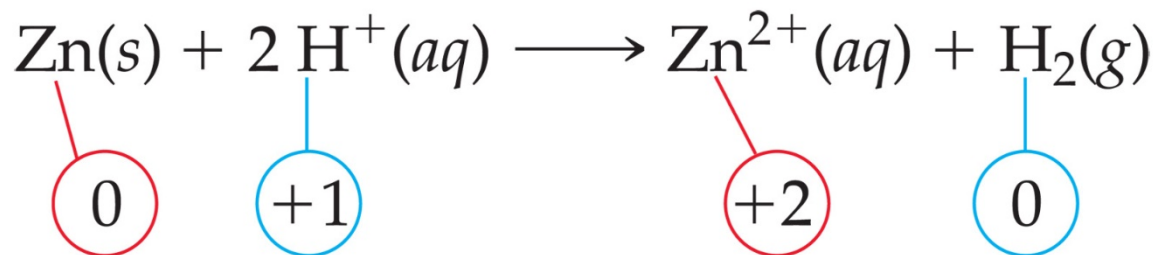
Reduction:

Half-Reaction:



Oxidation and Reduction

Zn metal reacting with HCl



- ✓ A species is **oxidized** when it **loses electrons**.
 - Here, zinc loses two electrons to go from neutral zinc metal to the Zn^{2+} ion.
- ✓ A species is **reduced** when it gains electrons.
 - Here, each of the H^+ gains an electron and they combine to form H_2 .
- ✓ What is reduced is the **oxidizing agent**.
 - H^+ oxidizes Zn by taking electrons from it.
- ✓ What is oxidized is the **reducing agent**.
 - Zn reduces H^+ by giving it electrons.

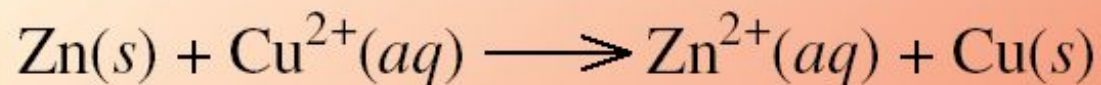
Electrochemistry

❖ The study of chemical reactions that produce electricity, and chemical reactions that occur due to electricity.

❖ There are many processes that take place because of electrochemical rxns:

- Electroplating
- Electrolysis of water
- Production of aluminum metal
- Storage of electricity in batteries

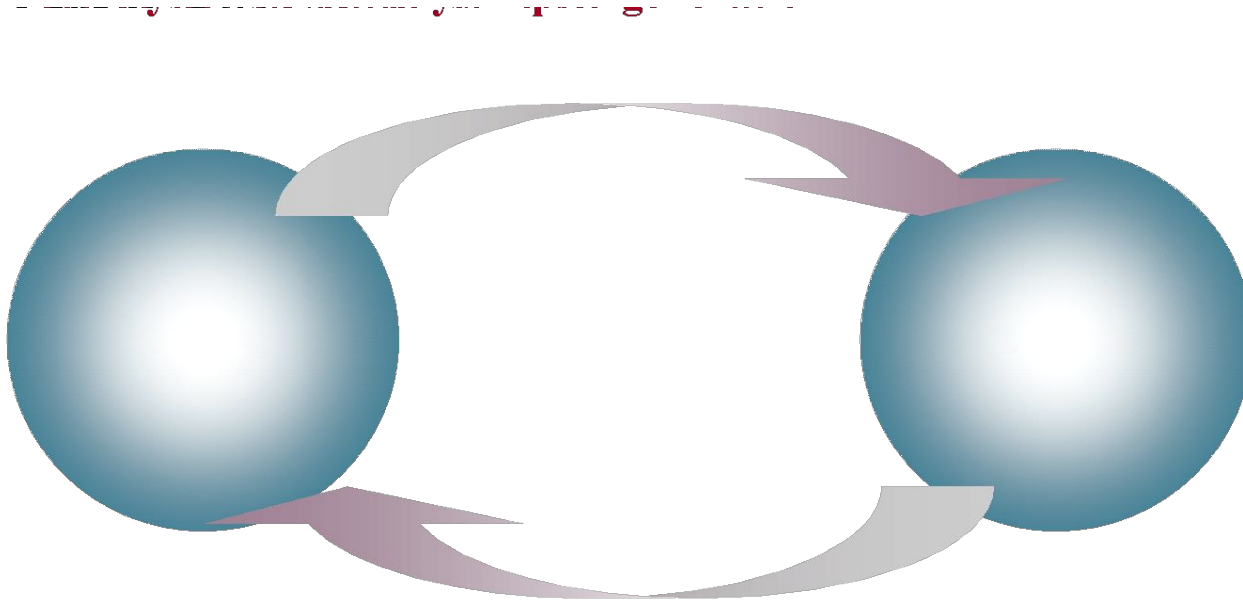
Electrochemical Cells



- Any electrochemical cell must consist of two electrodes.
- The current can flow only between two electrodes with different potentials.
- Every electrode is immersed in an electrolytic solution.
- The two electrolytic solutions are separated by porous separator.
- The electrodes ends, which not exposed to the electrolyte, are connected with each other by a metallic conductor.
- The current will be carried by electrons in the external conductors, whereas, ions carry it in the electrolytic solution.

Electrochemical Cells

There are two main types of electrochemical cells according the direction of the electric current.



- One of these cells is used in production of electric current by chemical reactions.

Galvanic cell.

- The other type of cells can produce chemical reaction using an electric current.

Electrolytic cell.

Electrolytic cell Vs Electrochemical cell:

An **electrochemical cell** uses a spontaneous chemical reaction to *generate* electricity, while an **electrolytic cell** uses electricity to *force a non-spontaneous chemical reaction* to occur.

Think of it like a battery vs. a battery charger.

Electrochemical Cell = A Battery (Discharging): It releases stored chemical energy as electrical energy.

•**Electrolytic Cell = A Battery Charger:** It uses electrical energy to store chemical energy.

Key Similarities between Electrolytic cell Vs Electrochemical cell

It's important to note what they have in common:

- Both involve redox reactions.
- Oxidation always occurs at the anode.
- Reduction always occurs at the cathode.
- Both contain an electrolyte to allow ion flow.
- Cations always move toward the cathode, and anions always move toward the anode.

Oxidation always happens at the anode , reduction always happens at the cathode, that is independent if cell is Electrolytic cell or Electrochemical cell. The electrons are always going to be moving from the anode to the cathode. If the cell is spontaneous, as galvanic is, it makes sense that the electrons will be moving towards positive electrode, the cathode is positive while for electrolytic, which is not spontaneous, an electron is actually being forced to move towards a negative pole ,hence the cathode is negative.

In the voltaic cell, the current flows from positive electrode (+) to the negative electrode (-) outside the cell.

In the electrolytic cell the currents flows from positive electrode to the negative electrode inside the cell.

Voltaic cell can become an electrolytic cell by use of an external voltage greater than the voltaic cell voltage and imposed in opposite direction.

Difference between Electrolytic cell and Electrochemical cell:

Feature	Electrochemical Cell (Galvanic or Voltaic Cell)	Electrolytic Cell
Definition	Converts chemical energy into electrical energy.	Converts electrical energy into chemical energy.
Reaction Type	Spontaneous ($\Delta G < 0$).	Non-spontaneous ($\Delta G > 0$).
Anode & Cathode	Anode: Negative electrode (source of electrons). Cathode: Positive electrode (sink for electrons).	Anode: Positive electrode (connected to + terminal of battery). Cathode: Negative electrode (connected to - terminal of battery).
Energy Conversion	Chemical $\rightarrow$ Electrical	Electrical $\rightarrow$ Chemical
External Power Source	Not required. The reaction provides the power.	Required. An external battery or power supply is essential.
Purpose/Use	To generate electrical power.	To drive a desired chemical reaction (electrolysis).
Examples	Batteries (AA, Li-ion), Fuel Cells , Daniel cell .	Electroplating (e.g., silver plating a spoon), Electrorefining of metals, Electrolysis of Water to produce H_2 and O_2 .
Oxidation & Reduction	Oxidation occurs at the anode. Reduction occurs at the cathode.	Oxidation occurs at the anode. Reduction occurs at the cathode.
Ion Flow in Electrolyte	Anions (negative) migrate to the anode.	Anions (negative) migrate to the anode.

Electrolytic Cell

Example:

Non-Spontaneous process

Imagine two inert Platinum (Pt) electrodes immersed in molten Sodium Chloride (NaCl), connected to a battery.

- **Process:** The battery "pulls" electrons from one electrode (making it the **anode**, positive) and "pushes" electrons onto the other (making it the **cathode**, negative).
 - At the **cathode (negative)**: Na^+ ions are attracted and gain electrons (reduction: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$).
 - At the **anode (positive)**: Cl^- ions are attracted and lose electrons (oxidation: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$).
- **Result:** A non-spontaneous reaction is forced to happen, decomposing NaCl into sodium metal and chlorine gas. Electrical energy is consumed.

ELECTROCHEMICAL CELLS/GALVANIC CELL:

An Electrochemical cell is a device in which a redox reaction is utilized to get electrical energy.

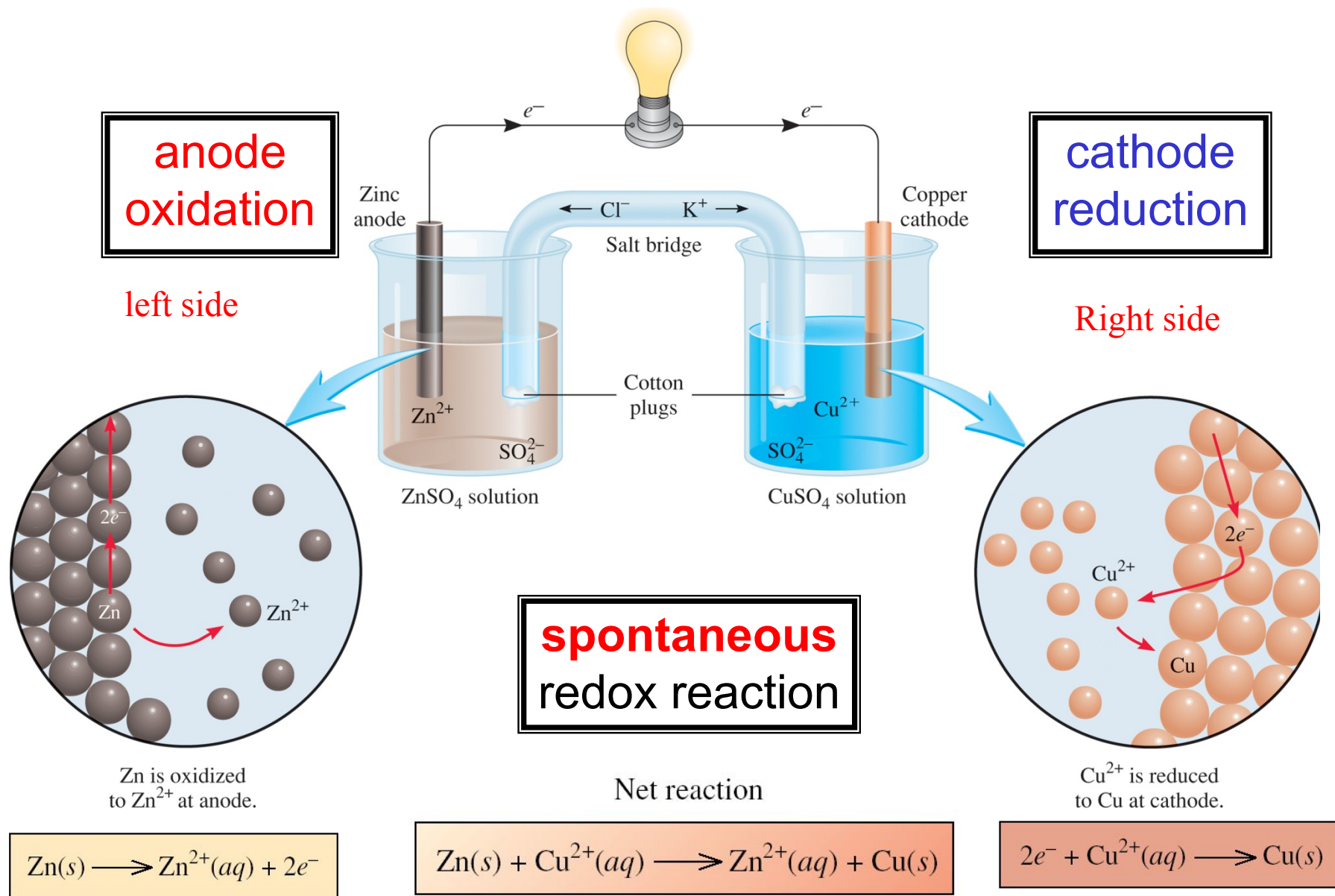
Spontaneous process

Ex: Daniel cell

Daniel cell consists of zinc and copper electrodes. Zn electrode is dipped in ZnSO₄ solution & Cu is dipped in CuSO₄ solution connected by a wire and a salt bridge.

- Process:** The more reactive Zn metal readily loses electrons (oxidizes: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$). These electrons flow through the wire to the Cu electrode, where Cu²⁺ ions gain them (reduce: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$).
- Result:** A spontaneous reaction occurs, creating a flow of electrons (electric current) that can power a light bulb. The anode (Zn) is negative, and the cathode (Cu) is positive.

Galvanic cells (also called Voltaic cells)



Galvanic Cells

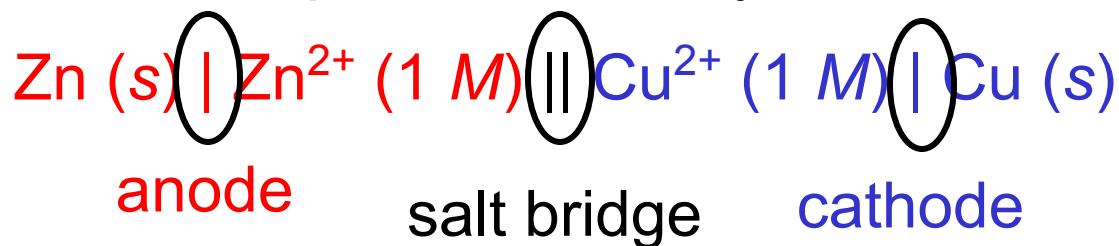
The difference in electrical potential between the anode and cathode is called:

- *cell voltage*
- *electromotive force (emf)*
- *cell potential*



Cell Diagram

phase boundary

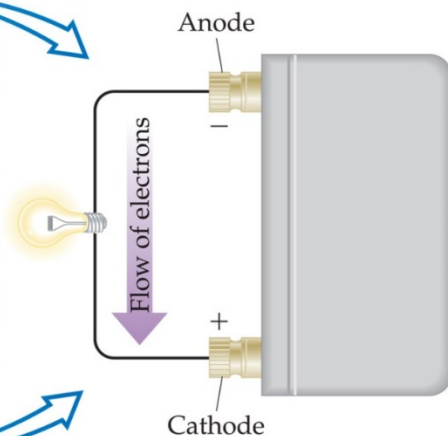


Electromotive Force (emf): Cell Potentials

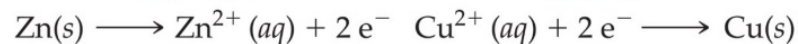
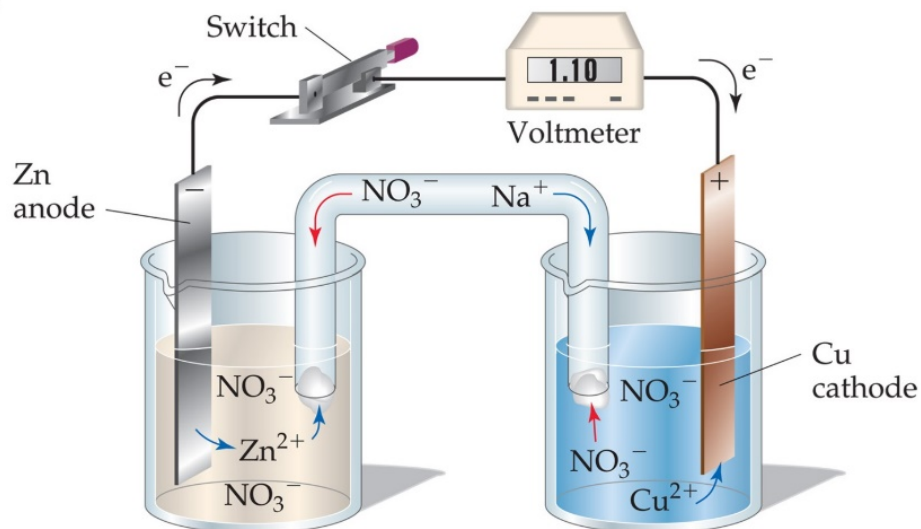


High
potential
energy

Low
potential
energy



- Water only spontaneously flows one way in a waterfall.
 - Likewise, electrons only spontaneously flow one way in a redox reaction—from higher to lower potential energy.



$$E = +1.10 \text{ V}$$

Cell reaction:

At Anode

Oxidation takes place: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$

At Cathode Reduction takes place: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$

The over all reaction is:



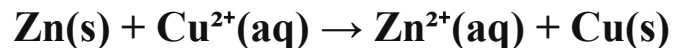
The two solutions separated by a porous membrane, a current is seen to be flow on connecting the two wires externally.

The cell function due to dissolution of zinc and the simultaneous deposition of copper.

Why are anode (Zn) negative and cathode (Cu) positive in Daniel cell?

1. The Core Process: Oxidation at the Anode

In a Daniell cell, the spontaneous reaction is:



At the **Zinc electrode (anode)**, oxidation occurs:



- Zinc atoms from the electrode *lose* two electrons and become Zn^{2+} ions that dissolve into the solution.
- These **two electrons are left behind on the Zinc electrode**.

2. Electron Buildup = Negative Charge

- The Zinc rod is made of neutral Zn atoms.
- As the reaction proceeds, it keeps pumping Zn^{2+} ions *into* the solution.
- For every Zn^{2+} ion that leaves the electrode, **two electrons are deposited onto the electrode itself**.

This continuous process causes a **buildup of electrons** on the Zinc electrode. An object with an excess of electrons acquires a **negative charge**.

Electron Drain = Positive Charge

Think of the copper electrode as a "destination" for electrons:

The copper rod is initially neutral.

As the reaction proceeds, Cu^{2+} ions in the solution are constantly *removing electrons* from the copper electrode to become Cu metal.

This creates a **deficit of electrons** on the copper electrode.

An object that has a deficit of electrons (i.e., has more protons than electrons) acquires a **positive charge**.

Why is anode positive and Cathode negative in electrolytic cell?

In an electrolytic cell, the electricity dictates the chemistry. We attach an external battery to force a non-spontaneous reaction.

- **The External Battery:** The battery has a fixed positive and negative terminal.
- **Anode Connection:** The anode of the electrolytic cell is physically connected to the **positive terminal** of the external battery.
- **Cathode Connection:** The cathode is connected to the **negative terminal**.

Why the Anode is Positive:

1. The positive terminal of the battery has a very high **electron deficiency**.
2. Since the anode is connected to it, the battery powerfully **pulls electrons away from the anode**.
3. This makes the anode electrode itself highly **positive**.
4. This strong positive charge then **attracts negative anions** from the solution.
5. When the anions reach the anode, the anode *forces* them to **lose electrons** (undergo oxidation). The pulled-away electrons are sent back to the battery.

Why the Cathode is Negative:

1. The negative terminal of the battery has a huge **excess of electrons**.
2. Since the cathode is connected to it, the battery **pushes electrons onto the cathode**.
3. This makes the cathode electrode itself highly **negative**.
4. This strong negative charge then **attracts positive cations** from the solution.
5. When the cations reach the cathode, the cathode *forces* them to **gain electrons** (undergo reduction).

3. The Role of the Circuit and the Cathode

- These excess electrons on the Zinc electrode have a strong repulsive force between them. They want to get away from each other.
- The external wire provides a path for them to do so. The electrons flow through the wire from the Zinc electrode (where they are in excess) to the Copper electrode (where they are in deficit).
- At the **Copper electrode (cathode)**, reduction occurs:
$$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$$
- The incoming electrons are *consumed* by the Cu^{2+} ions in the solution, plating solid copper onto the electrode.

4. Charge Separation and the Salt Bridge

This electron flow creates a charge imbalance:

- The **Zinc beaker** is becoming positively charged in the *solution* (due to the buildup of Zn^{2+} ions) and negatively charged on the *electrode*.
- The **Copper beaker** is becoming negatively charged in the *solution* (as Cu^{2+} ions are removed, leaving behind negative SO_4^{2-} ions) and positively charged on the *electrode* (as it loses electrons).

The **salt bridge** is crucial here. It allows ions to flow and neutralize this buildup, preventing the reaction from stopping almost immediately. It completes the circuit internally.

Uses of salt bridge:

- If you have no salt bridge, eventually all of the electrons from anode are going to move into the cathode and have zero electrons flow.
- It completes the circuit by connecting the electrodes at the electrode side.
- It maintains electrical charge neutrality in both the compartments during cell operation by selectively providing anions to anode and cations to cathode.

Cell Potentials

Calculation of the cell potential

The Standard, Universal Formula

The standard formula for calculating cell EMF (E°_{cell}) is:

$$E^\circ_{\text{cell}} = E^\circ(\text{reduction at cathode}) - E^\circ(\text{reduction at anode})$$

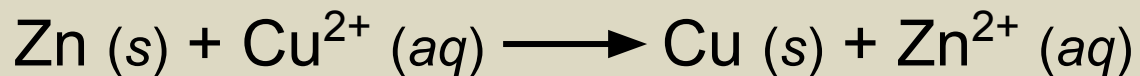
Example: Daniell Cell

For the Daniell cell:

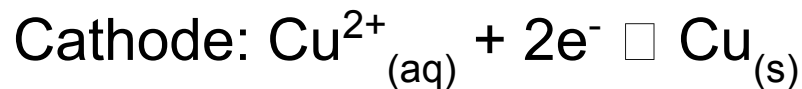
- **Cathode (Reduction):** $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu} \mid E^\circ = +0.34 \text{ V}$
- **Anode (Oxidation):** $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^- \mid E^\circ = -0.76 \text{ V}$ (but this is the reduction potential!)

Using the standard formula:

$$E^\circ_{\text{cell}} = E^\circ(\text{cathode}) - E^\circ(\text{anode}) = 0.34 \text{ V} - (-0.76 \text{ V}) = 1.10 \text{ V}$$

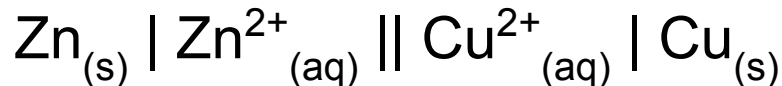


$$E_{\text{an}} = +0.76 \text{ V}$$



$$E_{\text{cat}} = +0.34 \text{ V}$$

The shorthand notation for this cell is:



Cell potentials

- ✓ The cell potential, E , is a measure of how well a cell reaction can push and pull electrons through a circuit
- ✓ The electrical energy generated by the spontaneous reaction is proportional to the cell potential.
- ✓ *The standard cell potential (the cell potential measured when all the species are in their standard states) is given by:*

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$$

EMF:

- In an electrochemical cell the potential difference across the electrodes operates as a driving force for the flow of electrons in one direction and thereby making current to flow in opposite direction and is known as cell potential/electromotive force(emf). It can be expressed in terms of both reduction potential and oxidation potential. In terms of tendency to gain electron, i.e., reduction potential of the cathode remains at higher potential. On the other hand, with respect to tendency to lose electron i.e., oxidation potential of the anode remains at higher potential

- In terms of reduction potential

- $$E^{\circ}_{\text{Cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}} \quad \text{or} \quad E^{\circ}_{\text{Right}} - E^{\circ}_{\text{Left}}$$

Commonly
represented

In terms of oxidation potential

- $$E^{\circ}_{\text{Cell}} = E^{\circ}_{\text{anode}} - E^{\circ}_{\text{cathode}}$$

Standard Cell Potentials (E°)

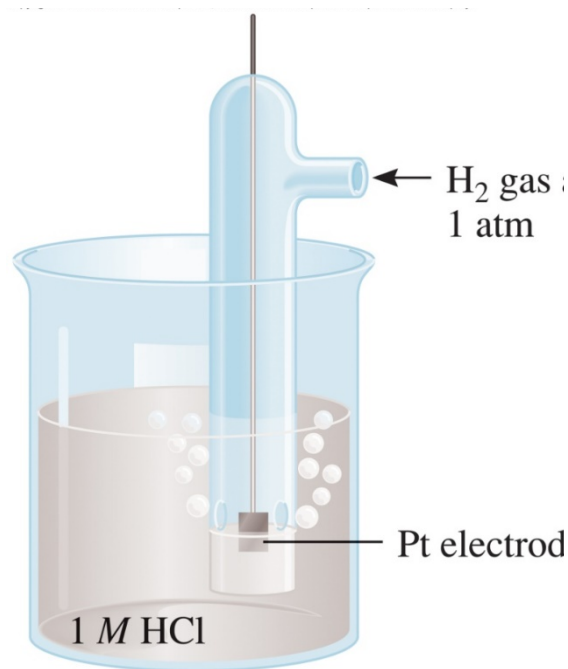
Standard Cell Potentials (E°)

- The cell potential at standard conditions can be found by looking up the reduction potentials for each half reaction on a table.
- The standard cell potentials (net E°) of all galvanic cells have positive values and happen spontaneously.
- The oxidation half reaction will be reversed from how it appears on the table, and the value of it's voltage will have a reversed sign too.

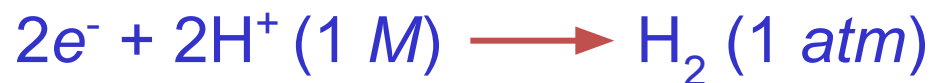
E° is a quantitative measure of the tendency of reactants to proceed to products when all are in their standard states at 25 °C.

Standard Reduction Potentials

- ❖ **Standard reduction potential (E^0)** is the voltage associated with a **reduction reaction** at an electrode when all solutes are 1 M and all gases are at 1 atm.



Reduction Reaction



$$E^0 = 0 V$$

- Can't measure half- reaction E^0 directly.
Therefore, measure it relative to a standard HALF CELL: the **Standard Hydrogen Electrode (SHE)**.

Standard hydrogen electrode (SHE)

SHE (H^+/H_2) gives us a universal reference point of zero, allowing us to compare all elements on the same scale.

Standard Hydrogen Electrode (SHE) - The Reference

This is the special case you mentioned, where platinum has a very specific role:

- Solution: 1 M H^+
- Gas: H_2 at 1 atm
- Reaction: $2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$
- By definition: $E^\circ = 0.000 \text{ V}$

Here, the platinum is:

- A catalyst for the H^+/H_2 reaction
- An electron conductor
- A surface for H_2 gas bubbles to form

Platinum is typically used as an **inert conductor** - its role is to provide a surface for electron transfer without participating in the redox reaction.

- While a reference point is necessary, the SHE's potential isn't inherently zero. Its potential is assigned a value of 0 V to facilitate electrochemistry.
- The SHE consists of a platinum electrode immersed in a 1 M H^+ solution and exposed to H_2 gas at 1 atm pressure

The **0.00 V** potential belongs to the H^+/H_2 couple, not the platinum.

Why Platinum is the Perfect Choice for the Hydrogen Electrode

The Standard Hydrogen Electrode is based on the reaction:



Platinum is ideal for facilitating this reaction because of these key properties:

1. Inertness (Chemical Stability)

- Platinum is a noble metal. It does not corrode or dissolve significantly in most acidic solutions.
- It doesn't react with the H^+ ions or H_2 gas, so it doesn't introduce its own side reactions that would alter the potential.

2. High Catalytic Activity

- This is the most important reason. The H^+/H_2 reaction is often slow on other surfaces.
- Platinum acts as an excellent **catalyst** for the interconversion between H^+ ions, electrons, and H_2 molecules. It provides a surface where the reaction:
$$2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2$$

can occur rapidly and reversibly, allowing the electrode to quickly reach its equilibrium potential.

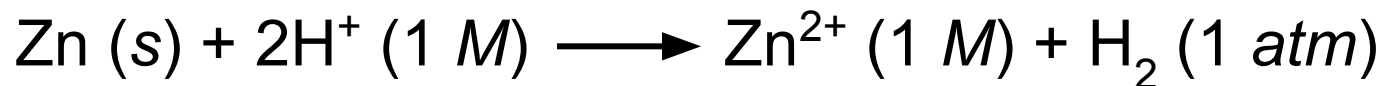
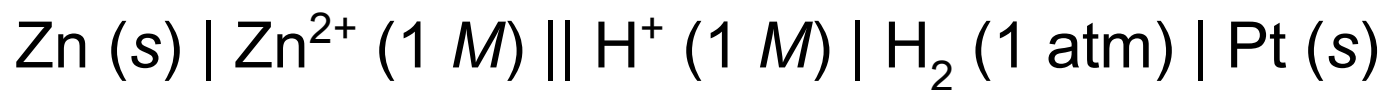
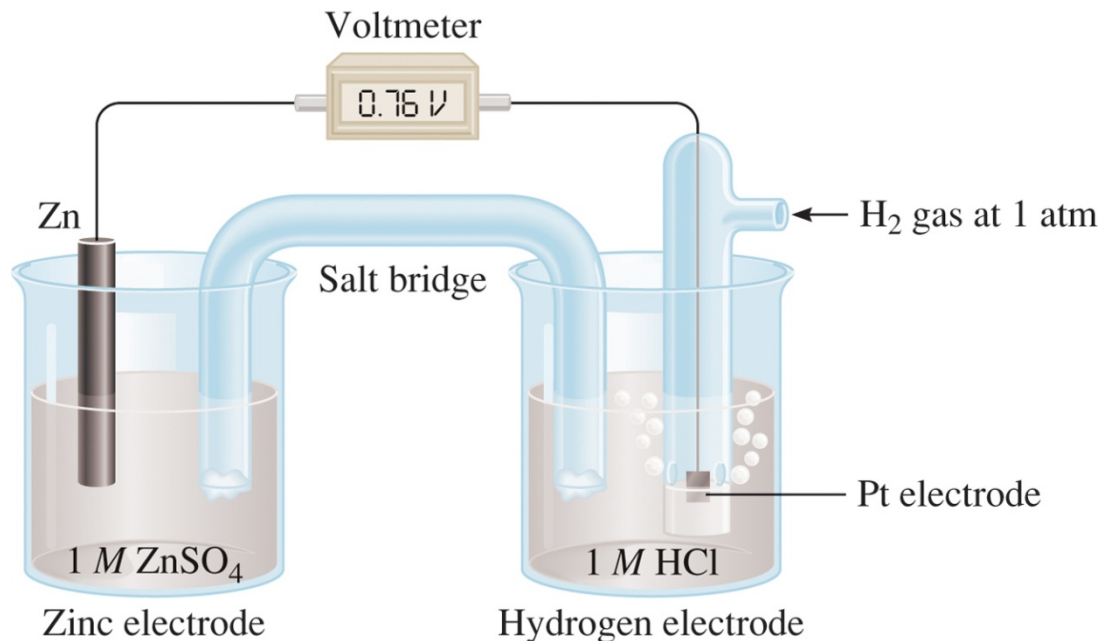
3. Electrical Conductivity

- As a metal, platinum is an excellent conductor of electrons, which is essential for any electrode.

4. High Exchange Current Density

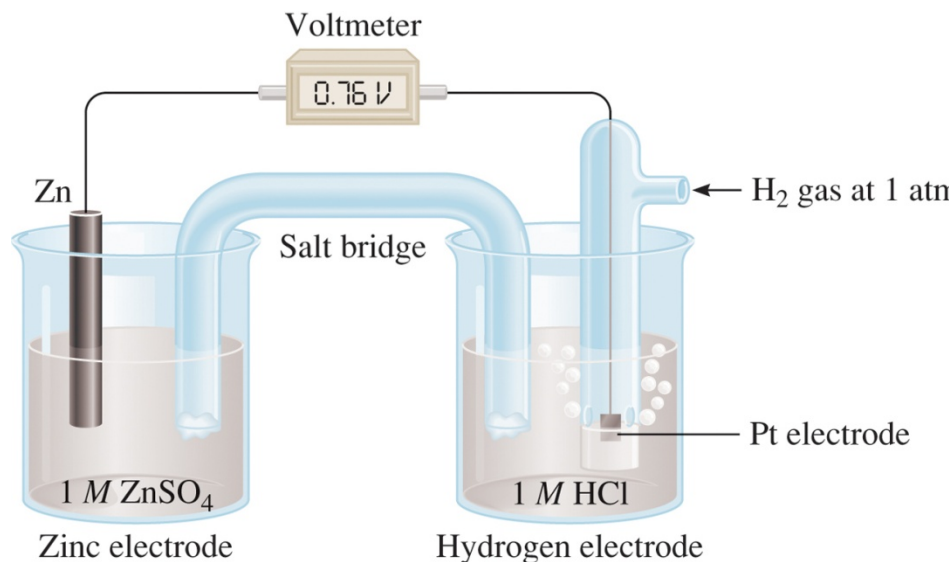
- This is an electrochemical way of saying that the equilibrium for the H^+/H_2 reaction is established very quickly and robustly on a platinum surface. The electrode potential is stable and doesn't get easily disturbed by small currents passing through it.

Standard Reduction Potentials of $\text{Zn (s)} \mid \text{Zn}^{2+} (1 \text{ M})$



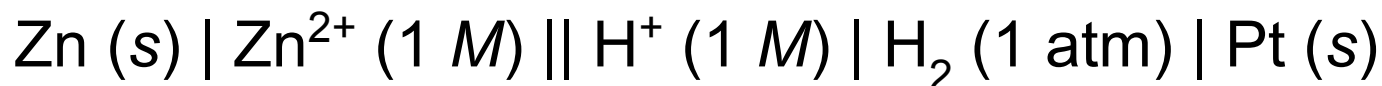
Standard Reduction Potentials

$$E_{\text{cell}}^0 = 0.76 \text{ V}$$



Standard emf (E_{cell}^0)

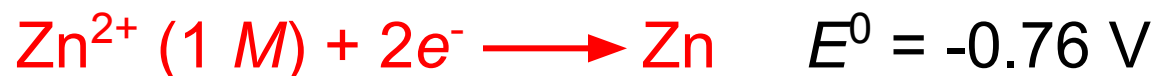
$$E_{\text{cell}}^0 = E_{\text{cathode}}^0 - E_{\text{anode}}^0$$



$$E_{\text{cell}}^0 = E_{\text{H}^+/\text{H}_2}^0 - E_{\text{Zn}^{2+}/\text{Zn}}^0$$

$$0.76 \text{ V} = 0 - E_{\text{Zn}^{2+}/\text{Zn}}^0$$

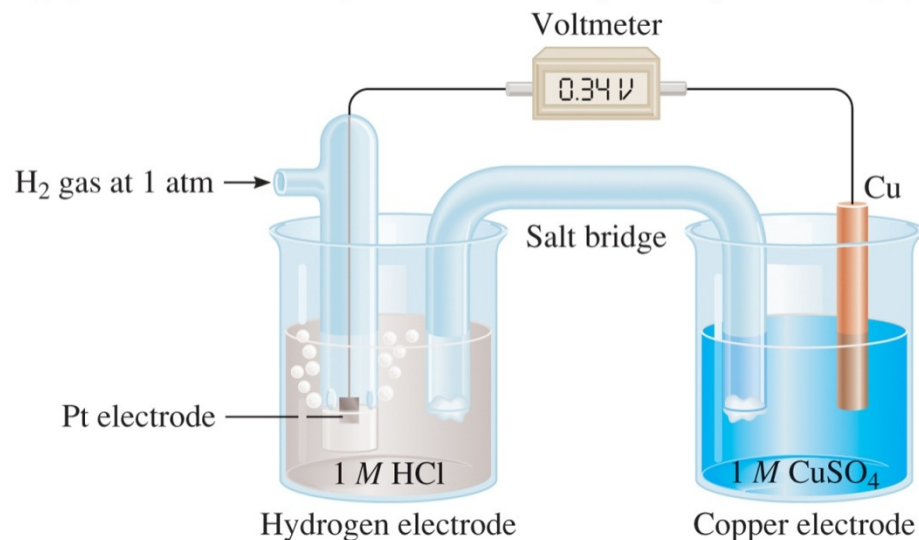
$$E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76 \text{ V}$$



Standard Reduction Potentials of $\text{Cu}^{2+} (1 M) | \text{Cu} (s)$

$$E_{\text{cell}}^0 = 0.34 \text{ V}$$

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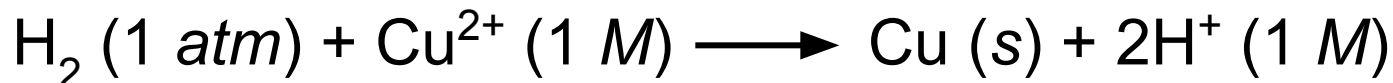
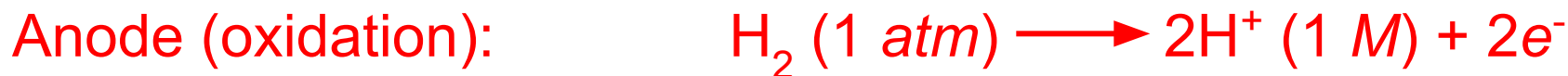
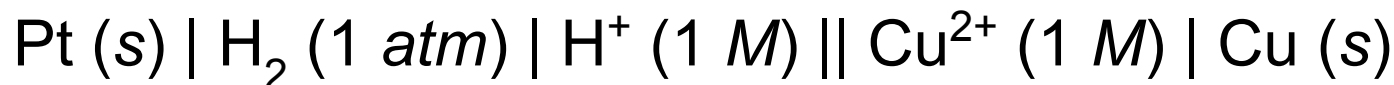


$$E_{\text{cell}}^0 = E_{\text{cathode}}^0 - E_{\text{anode}}^0$$

$$E_{\text{cell}}^0 = E_{\text{Cu}^{2+}/\text{Cu}}^0 - E_{\text{H}^+/\text{H}_2}^0$$

$$0.34 = E_{\text{Cu}^{2+}/\text{Cu}}^0 - 0$$

$$E_{\text{Cu}^{2+}/\text{Cu}}^0 = 0.34 \text{ V}$$



When measuring the standard reduction potential of $\text{Zn (s)} \mid \text{Zn}^{2+}$, how do we know that we must use Zn electrode as anode, where as for Cu^{2+}/Cu as cathode?

The short answer is: **We don't "know" beforehand. We define one half-cell as the reference point, and all other half-cells are measured relative to it.**

The Universal Reference Point: The Standard Hydrogen Electrode (SHE)

To solve the "we need a reference" problem, scientists agreed upon a universal standard: the **Standard Hydrogen Electrode (SHE)**.

- **Half-Cell:** $2\text{H}^+(\text{aq}, 1 \text{ M}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g}, 1 \text{ atm})$
- **Electrode:** An inert conductor like Platinum (Pt).
- **Assigned Potential:** $E^\circ = 0.000 \text{ V}$ by definition.

The SHE is the "sea level" for electrochemical measurements.

How We Measure the Potential for Zn^{2+}/Zn

Now, let's see how we determine the potential for the zinc half-cell.

1. Build the Cell: We connect the SHE to a Zn half-cell.

- 1. Half-Cell A (Reference):** SHE (H^+/H_2 on Pt)
- 2. Half-Cell B (To be measured):** $\text{Zn(s)} \mid \text{Zn}^{2+}(\text{aq}, 1 \text{ M})$

2. Observe and Measure:

- We observe that the voltmeter shows a positive voltage, and we see that the Zinc electrode is dissolving. Hydrogen gas is being produced at the platinum electrode in the SHE.
- This tells us that the spontaneous reaction in the cell is:
 - Anode (Oxidation):** $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$
 - Cathode (Reduction in SHE):** $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$
- The measured cell potential (E°_{cell}) is **+0.76 V**.

3. Assign the Value:

- We know the formula: **$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$**
- Plugging in our values: **$0.76 \text{ V} = 0.00 \text{ V} - E^\circ(\text{Zn}^{2+}/\text{Zn})$**
- Solving for the unknown: **$E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$**

Conclusion: We found out that Zn acts as the **anode** (the site of oxidation) when paired with the SHE. Therefore, its reduction potential is negative. We *had* to use a Zn electrode because that is the solid conductor through which electrons enter or leave the Zn^{2+} solution.

How We Measure the Potential for Cu^{2+}/Cu

Now, let's do the same for copper.

1. Build the Cell: We connect the SHE to a Cu half-cell.

- 1. Half-Cell A (Reference):** SHE (H^+/H_2 on Pt)
- 2. Half-Cell B (To be measured):** $\text{Cu(s)} \mid \text{Cu}^{2+}(\text{aq}, 1 \text{ M})$

2. Observe and Measure:

1. We observe that the voltmeter shows a positive voltage, but now we see copper metal depositing onto the copper electrode. The H_2 gas in the SHE is being consumed.
2. This tells us that the spontaneous reaction in the cell is:
 - 1. Anode (Oxidation in SHE):** $\text{H}_2(\text{g}) \rightarrow 2\text{H}^+(\text{aq}) + 2\text{e}^-$
 - 2. Cathode (Reduction):** $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$
3. The measured cell potential (E°_{cell}) is **+0.34 V**.

Assign the Value:

Using the formula: $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$

Plugging in our values: **$0.34 \text{ V} = E^\circ(\text{Cu}^{2+}/\text{Cu}) - 0.00 \text{ V}$**

Solving for the unknown: **$E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$**

Conclusion: We found out that Cu acts as the **cathode** (the site of reduction) when paired with the SHE. Therefore, its reduction potential is positive. Again, we *had* to use a Cu electrode for the same reason—it's the necessary conductor for the Cu^{2+}/Cu half-reaction.

Why are we measuring Standard Reduction Potentials of $\text{Cu}^{2+} (1 \text{ M}) | \text{Cu} (s)$ and $\text{Zn} (s) | \text{Zn}^{2+}$ with respect to the H^+/H_2 ?

Voltage is a Difference, Not an Absolute Value

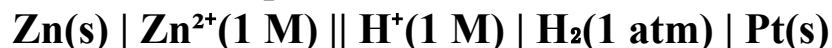
A crucial concept is that **you can never measure an absolute "single electrode potential."** You can only measure the *difference in potential* between two electrodes (the EMF or voltage of a cell).

Think of altitude: You can't say "Mount Everest is 8848 meters" in an absolute sense. That measurement is *relative to* the average sea level, which we've arbitrarily defined as 0 meters.

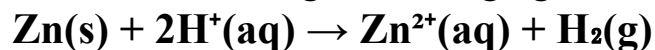
In electrochemistry, the "sea level" is the **Standard Hydrogen Electrode (SHE)**.

2. What Are We Actually Measuring?

When we set up this cell:



We are measuring the voltage generated by the spontaneous reaction:



The measured voltage (about 0.76 V) is the **difference in tendency** between the two half-cells to gain electrons.

The Zn half-cell has a strong tendency to *lose* electrons (be oxidized).

The H^+/H_2 half-cell has a weaker tendency to *lose* electrons.

Since we've defined the H^+/H_2 tendency as "zero," the entire measured voltage is assigned to the Zn half-cell.

So, **E° for Zn^{2+}/Zn is -0.76 V**

Standard Reduction Potentials

The species with the more positive (V) will be reduced.

The species lowest on the table, and the more negative (V) will be oxidized.

Potential (V)	Reduction Half-Reaction
+2.87	$\text{F}_2(\text{g}) + 2 \text{e}^- \longrightarrow 2 \text{F}^-(\text{aq})$
+1.51	$\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 5 \text{e}^- \longrightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\text{l})$
+1.36	$\text{Cl}_2(\text{g}) + 2 \text{e}^- \longrightarrow 2 \text{Cl}^-(\text{aq})$
+1.33	$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14 \text{H}^+(\text{aq}) + 6 \text{e}^- \longrightarrow 2 \text{Cr}^{3+}(\text{aq}) + 7 \text{H}_2\text{O}(\text{l})$
+1.23	$\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 \text{e}^- \longrightarrow 2 \text{H}_2\text{O}(\text{l})$
+1.06	$\text{Br}_2(\text{l}) + 2 \text{e}^- \longrightarrow 2 \text{Br}^-(\text{aq})$
+0.96	$\text{NO}_3^-(\text{aq}) + 4 \text{H}^+(\text{aq}) + 3 \text{e}^- \longrightarrow \text{NO}(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$
+0.80	$\text{Ag}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Ag}(\text{s})$
+0.77	$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \longrightarrow \text{Fe}^{2+}(\text{aq})$
+0.68	$\text{O}_2(\text{g}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \longrightarrow \text{H}_2\text{O}_2(\text{aq})$
+0.59	$\text{MnO}_4^-(\text{aq}) + 2 \text{H}_2\text{O}(\text{l}) + 3 \text{e}^- \longrightarrow \text{MnO}_2(\text{s}) + 4 \text{OH}^-(\text{aq})$
+0.54	$\text{I}_2(\text{s}) + 2 \text{e}^- \longrightarrow 2 \text{I}^-(\text{aq})$
+0.40	$\text{O}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l}) + 4 \text{e}^- \longrightarrow 4 \text{OH}^-(\text{aq})$
+0.34	$\text{Cu}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Cu}(\text{s})$
0 [defined]	$2 \text{H}^+(\text{aq}) + 2 \text{e}^- \longrightarrow \text{H}_2(\text{g})$
-0.28	$\text{Ni}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Ni}(\text{s})$
-0.44	$\text{Fe}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Fe}(\text{s})$
-0.76	$\text{Zn}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Zn}(\text{s})$
-0.83	$2 \text{H}_2\text{O}(\text{l}) + 2 \text{e}^- \longrightarrow \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$
-1.66	$\text{Al}^{3+}(\text{aq}) + 3 \text{e}^- \longrightarrow \text{Al}(\text{s})$
-2.71	$\text{Na}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Na}(\text{s})$
-3.05	$\text{Li}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Li}(\text{s})$

Standard Reduction Potentials

- ✓ *The more positive E^0 the greater the tendency for the substance to be reduced*
- ✓ The half-cell reactions are reversible.
- ✓ The sign of E^0 changes when the reaction is reversed
- ✓ Changing the stoichiometric coefficients of a half-cell reaction *does not* change the value of E^0

Comparison Table of Common Reference Electrodes

Electrode	Abbreviation	Potential vs. SHE (at 25°C)	Common Use Cases
Standard Hydrogen	SHE	0.000 V (by definition)	Primary standard; theoretical reference
Saturated Calomel	SCE	+0.241 V	Classic laboratory work (being phased out)
Silver/Silver Chloride	Ag/AgCl	+0.197 V (sat. KCl)	Modern laboratories, biology, medicine, pH electrodes
Copper/Copper Sulfate	CSE	+0.318 V	Corrosion monitoring, field measurement.

How to Use Them in Practice

When you use one of these alternative electrodes, you simply **add its potential** to your measured cell voltage to find the value relative to the SHE.

Formula:

$$E(\text{vs. SHE}) = E(\text{measured}) + E(\text{reference electrode})$$

Example:

You are using a Saturated Calomel Electrode (SCE) to measure the potential of a zinc half-cell.

1. You connect your cell: $\text{Zn} \mid \text{Zn}^{2+} \parallel \text{SCE}$
2. Your voltmeter reads: **$E(\text{measured}) = -1.001 \text{ V}$**
3. You know: **$E(\text{SCE vs. SHE}) = +0.241 \text{ V}$**
4. Calculate the standard potential:
 - $E(\text{Zn vs. SHE}) = -1.001 \text{ V} + 0.241 \text{ V} = \textbf{-0.760 V}$

This gives you the standard value you can look up in a table.

In short:

- **SHE is the "Gold Standard":** It defines the zero point (like Greenwich Mean Time for time zones).
 - **Other Electrodes are "Local Time Zones":** They are much more convenient for daily use. SCE, Ag/AgCl, and CSE are like EST, PST, etc.—they have a known, fixed offset from the prime standard.
 - **Choice Depends on Application:** The choice of reference electrode depends on the required precision, chemical compatibility, temperature, and whether you're in a lab or the field.
- So, while the SHE is fundamental to the *definition* of electrode potentials, you will most often encounter Ag/AgCl or SCE electrodes on the lab bench.

Relation of E° and ΔG°

E° is related to ΔG° , the free energy change for the reaction.

$$\Delta G^\circ = -n F E^\circ$$

- F = Faraday constant = $9.6485 \times 10^4 \text{ J/V}\cdot\text{mol}$
- n = the number of moles of electrons transferred.



Michael Faraday
1791-1867

Discoverer of

- electrolysis
- magnetic props. of matter
- electromagnetic induction
- benzene and other organic chemicals



n for Zn/Cu cell ?

$$n = 2$$

Nernst Equation

□ Have relationship between Gibbs Free Energy and Equilibrium constant:

$$\Delta G^\circ = -RT \ln K$$

□ ΔG for a reaction depends on the concentration by:

$$\Delta G = \Delta G^\circ + RT \ln Q \quad \text{where } Q \text{ is the reaction quotient} = \frac{[\text{product}]}{[\text{reactant}]}$$

$$\Delta G = -n F E_{\text{cell}} \quad \text{and} \quad \Delta G^\circ = -n F E_{\text{cell}}^\circ$$

$$-n F E_{\text{cell}} = -n F E_{\text{cell}}^\circ + RT \ln Q$$

Dividing across by nF gives:

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \ln Q$$

Nernst Equation

i.e. the cell potential at any conditions depends on the potential under standard state conditions and a term for the potential at nonstandard-state conditions